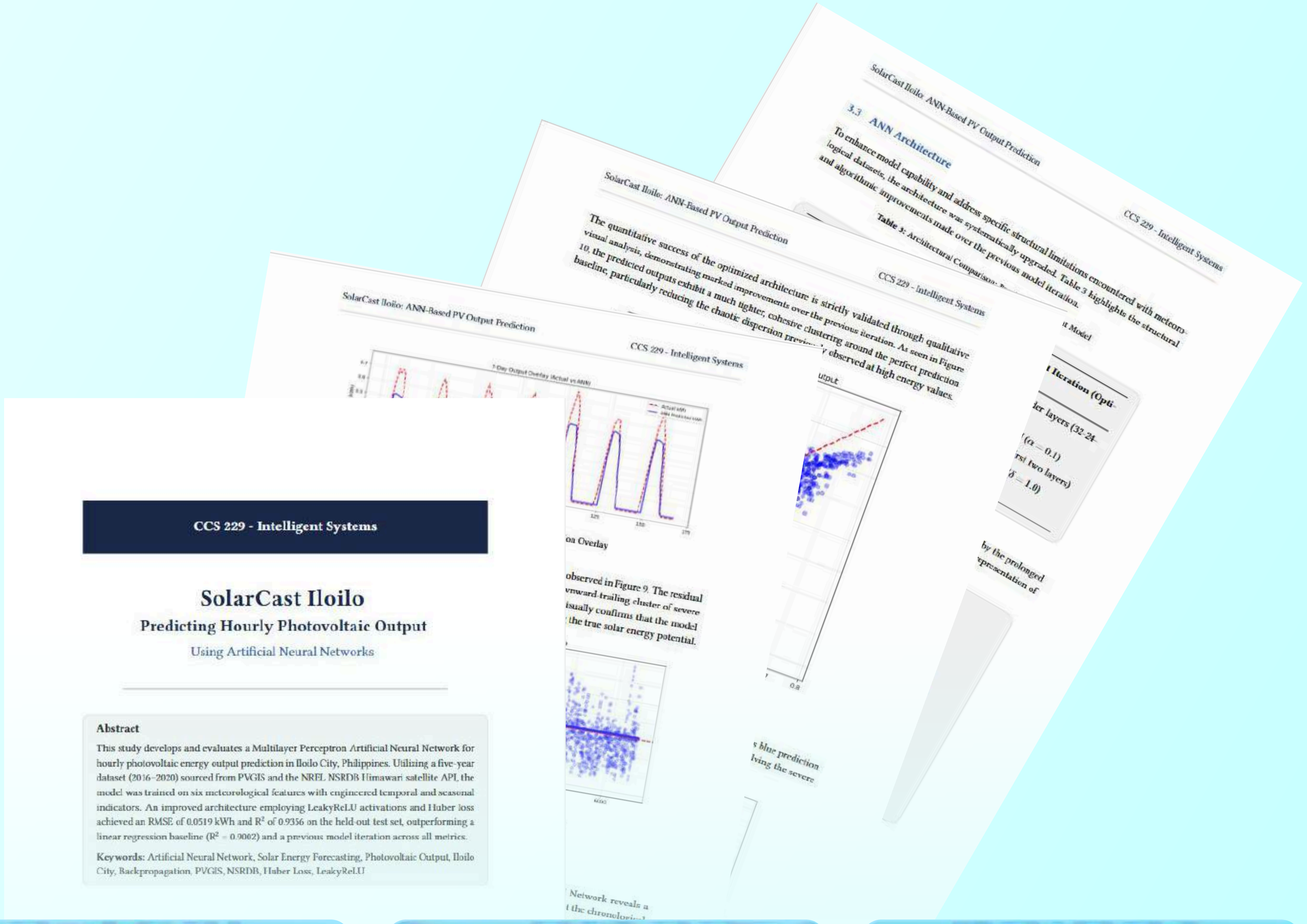
SOLARCAST ILOILO

Predicting Hourly Photovoltaic Output
Using Artificial Neural Networks



5-Year
DATASET

R^2 0.9356
MODEL ACCURACY

43,839
Hourly Records

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INTRODUCTION

THE CHALLENGE

Solar power is inherently intermittent — its output depends on rapidly changing meteorological conditions.

Without accurate forecasts, grid operators struggle to balance supply and demand in the Visayas WESM, leading to:

- **Economic inefficiencies**
- **Grid instability risks**
- **Sub-optimal dispatch scheduling**

OBJECTIVE

Can a Multilayer Perceptron ANN accurately predict hourly PV output from localized weather data?

Scope:

- Location: Iloilo City (10.7019°N, 122.5622°E)
- Reference system: 1 kWp photovoltaic array
- Dataset: 2016–2020 (5-year hourly)
- Comparison: ANN vs. Linear Regression vs. Previous Model

DATASET & DATA SOURCES

PVGIS-ERA5

Target Variable

Estimated hourly PV power output

Converted to energy_kwh

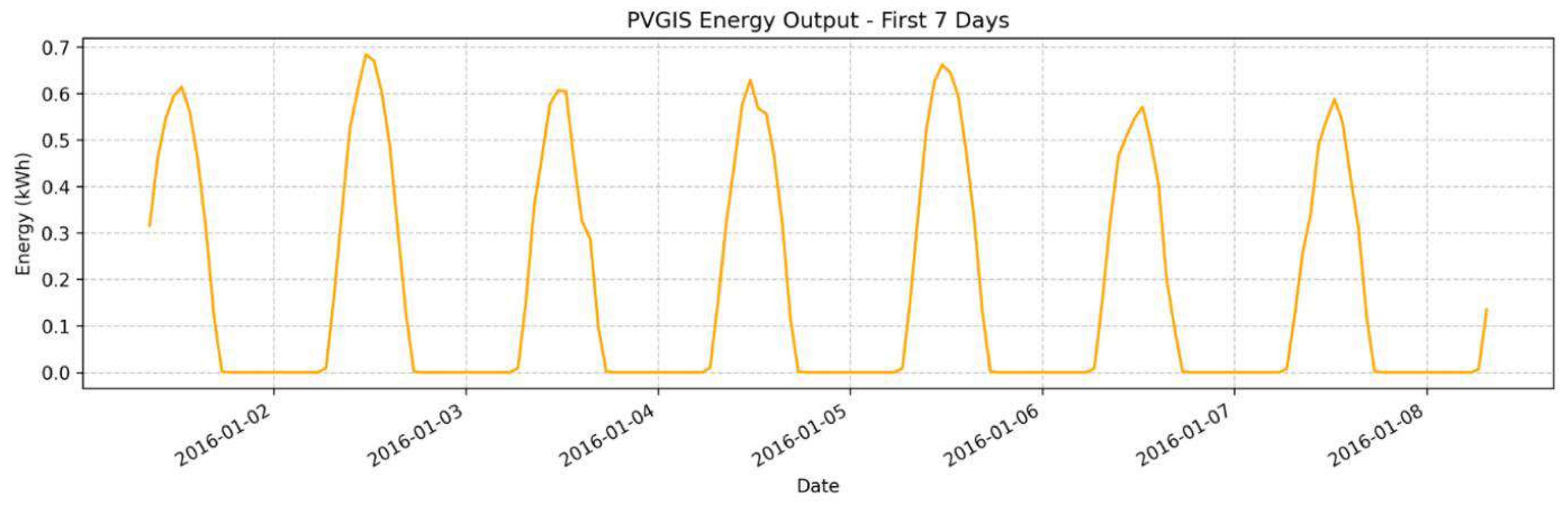

Figure 1: Diurnal Solar Generation Pattern

NREL NSRDB

Meteorological Inputs (Himawari Satellite)

- Global Horizontal Irradiance (GHI)
- Direct Normal Irradiance (DNI)
- Diffuse Horizontal Irradiance (DHI)
- Temperature, Wind Speed, Relative Humidity

5-Year
2016 – 2020

6
Meteorological Features

43,839
Hourly Records

DATA PREPROCESSING

Timezone Alignment	PVGIS (UTC) shifted +8h to match NSRDB Philippine Standard Time
Dataset Merge	Inner join on synchronized datetime index → 43,839 clean hourly records
Feature Engineering	Hour, month, is_wet_season flag (Jun–Nov = typhoon/monsoon season)
Chronological Split	Train 70% (30,687) / Validation 15% (6,576) / Test 15% (6,576)
Normalization	MinMaxScaler on continuous features, fit on training set only (no leakage)

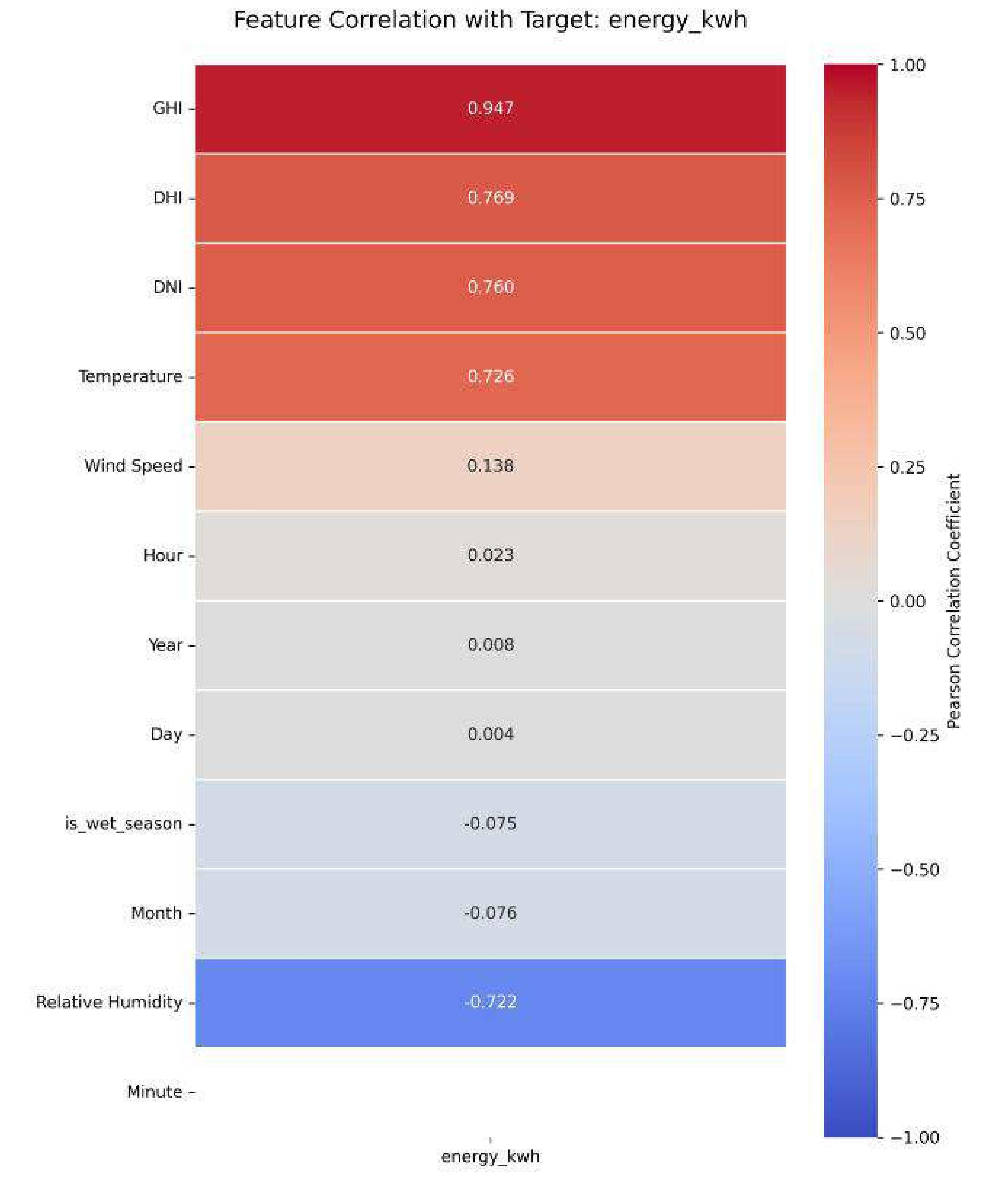

Figure 2: Feature Correlation Matrix

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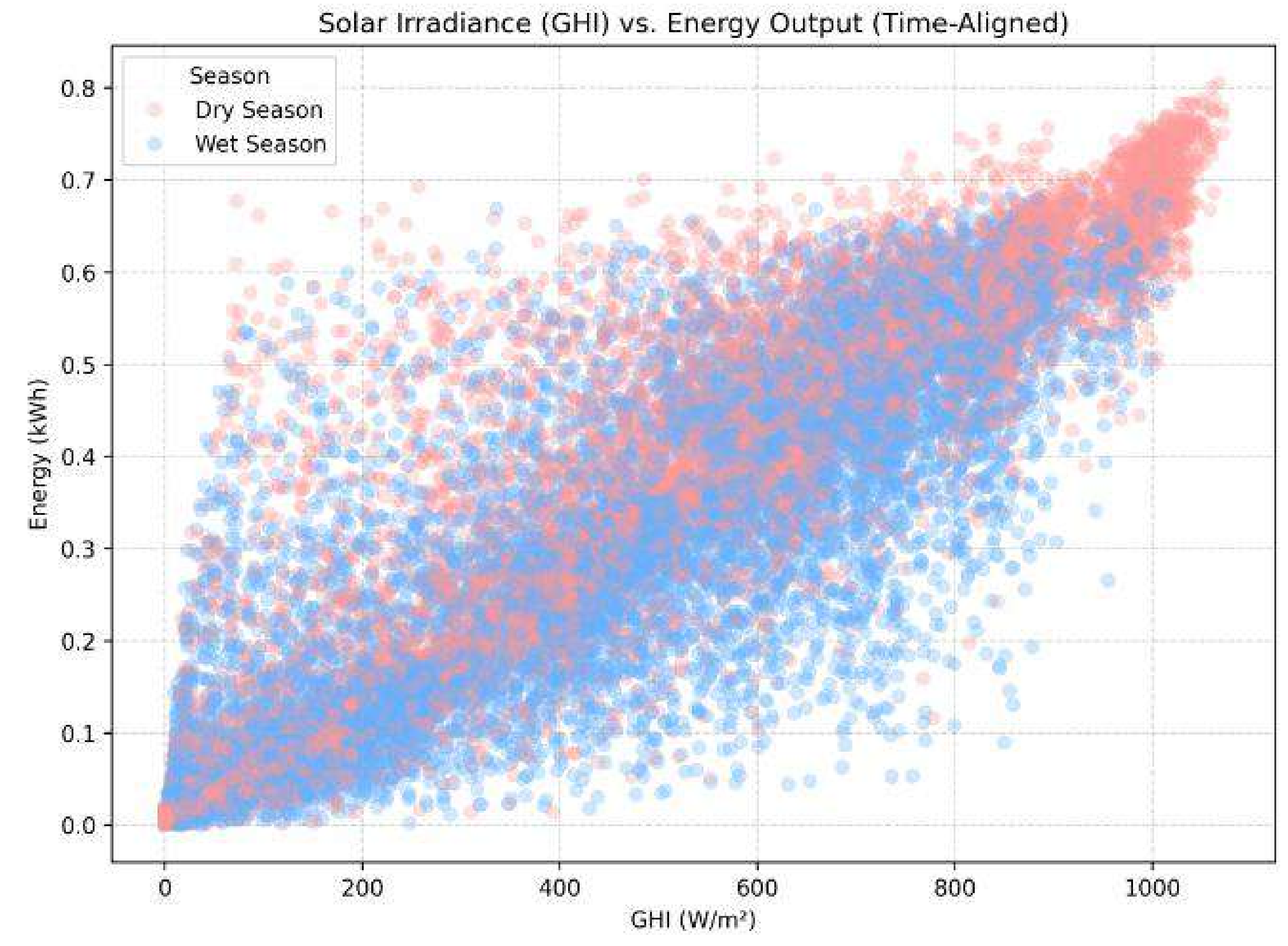
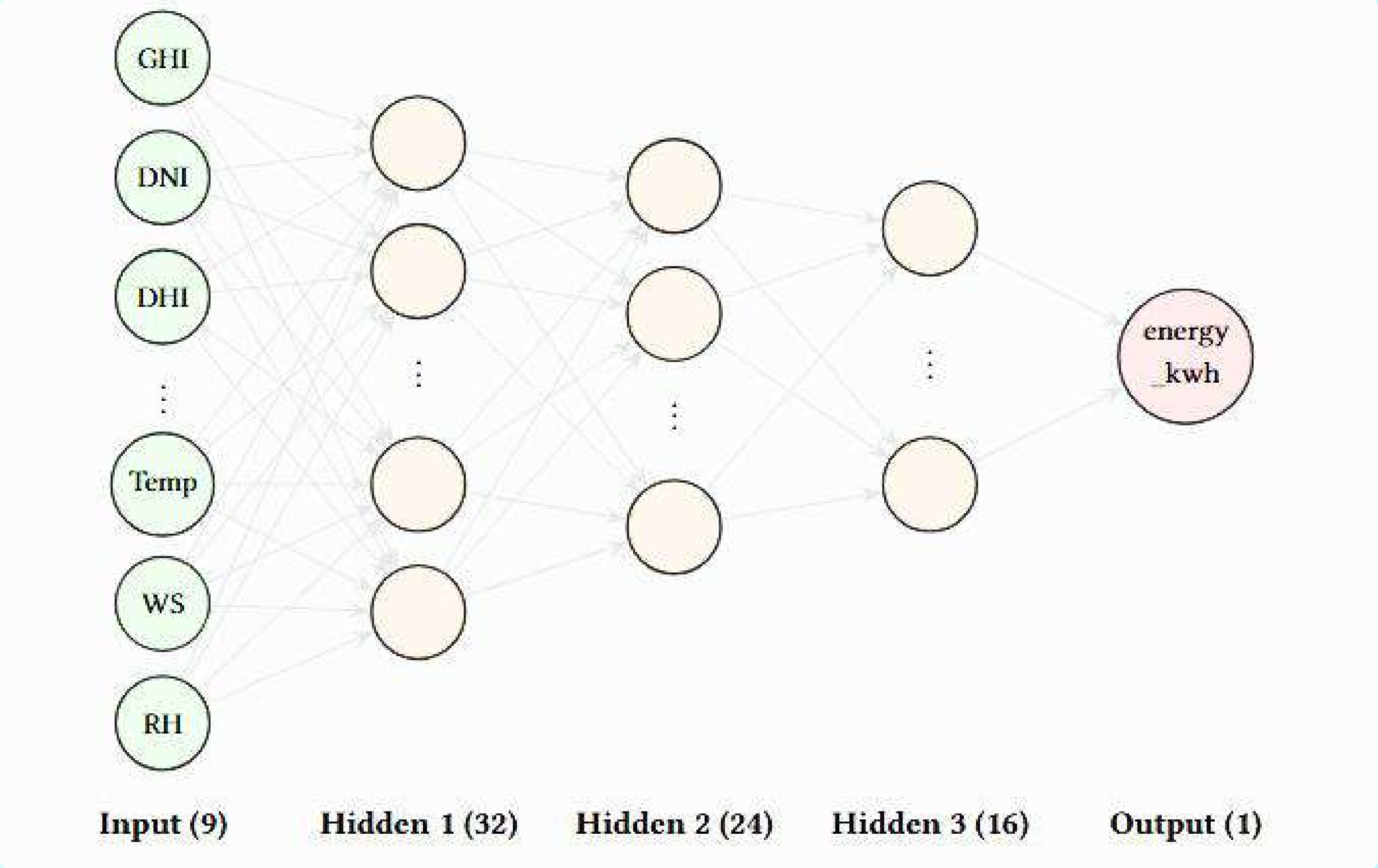


Figure 3: GHI vs. Energy Output Scatter Plot with Seasonal Differentiation

ANN ARCHITECTURE

Table 3: Architectural Comparison: Previous vs. Current Model

Parameter	Previous Iteration	Current Iteration (Optimized)
Hidden Layers (Neurons)	Three (16-12-8)	Three wider layers (32-24-16)
Activation Function	ReLU	LeakyReLU ($\alpha = 0.1$)
Dropout Regularization	0.20 (after first two layers)	0.15 (after first two layers)
Loss Function	Mean Squared Error (MSE)	Huber Loss ($\delta = 1.0$)
Total Trainable Params	Not Recorded	1,529



MODEL TRAINING

OPTIMIZER	BATCH SIZE	MAX EPOCHS	BEST EPOCH
ADAM	64	200	42
LR: 0.001		Stopped at 57	Weights restored

Keras Callbacks

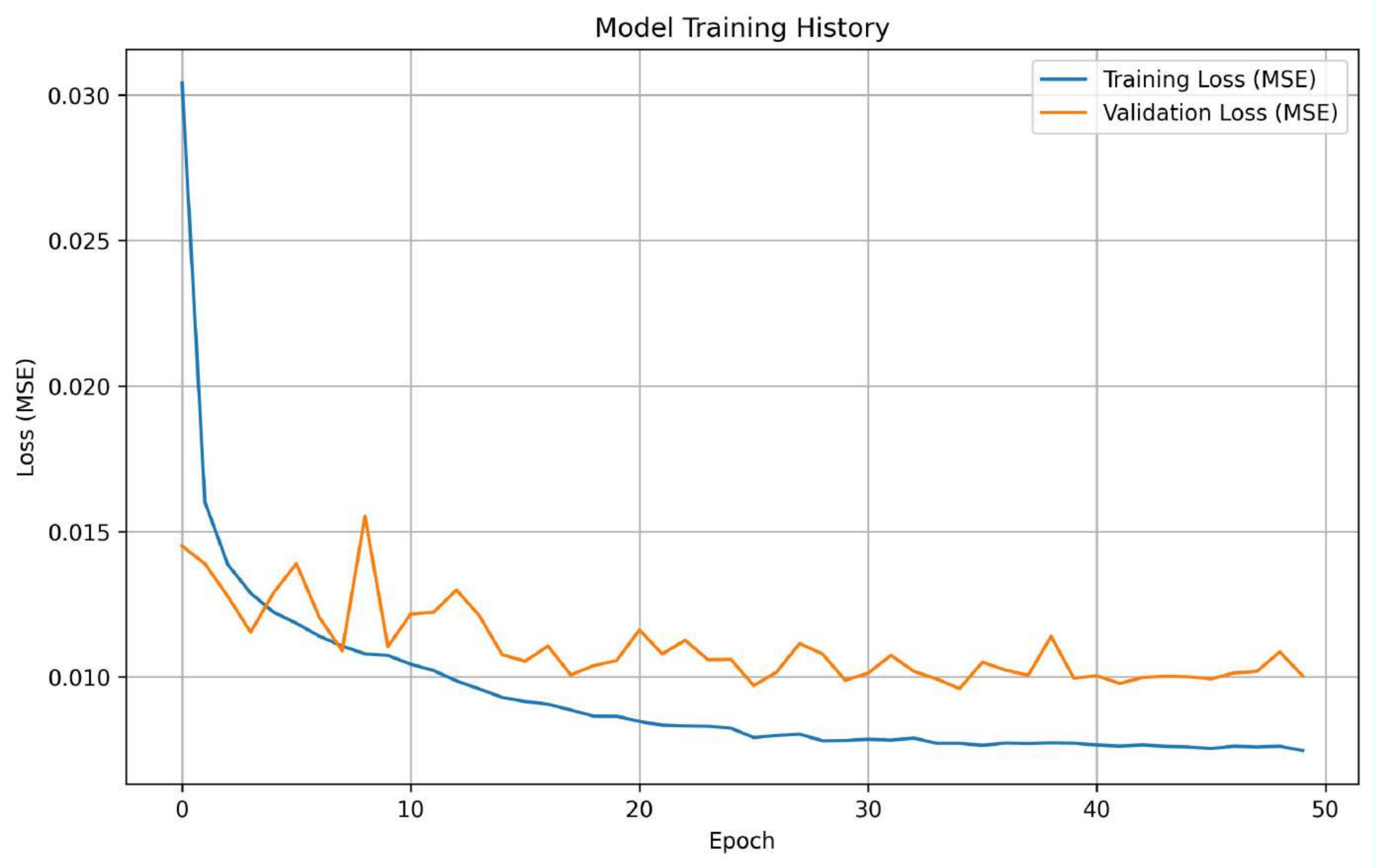
Early Stopping	Monitors val_loss; stops after 15 stagnant epochs; restores best weights
ModelCheckpoint	Serializes best model iteration to models/best_model_iloilo.keras
ReduceLROnPlateau	Halves learning rate (×0.5) if val_loss stalls 7 epochs; min LR = 1e-6

Best Epoch Metrics

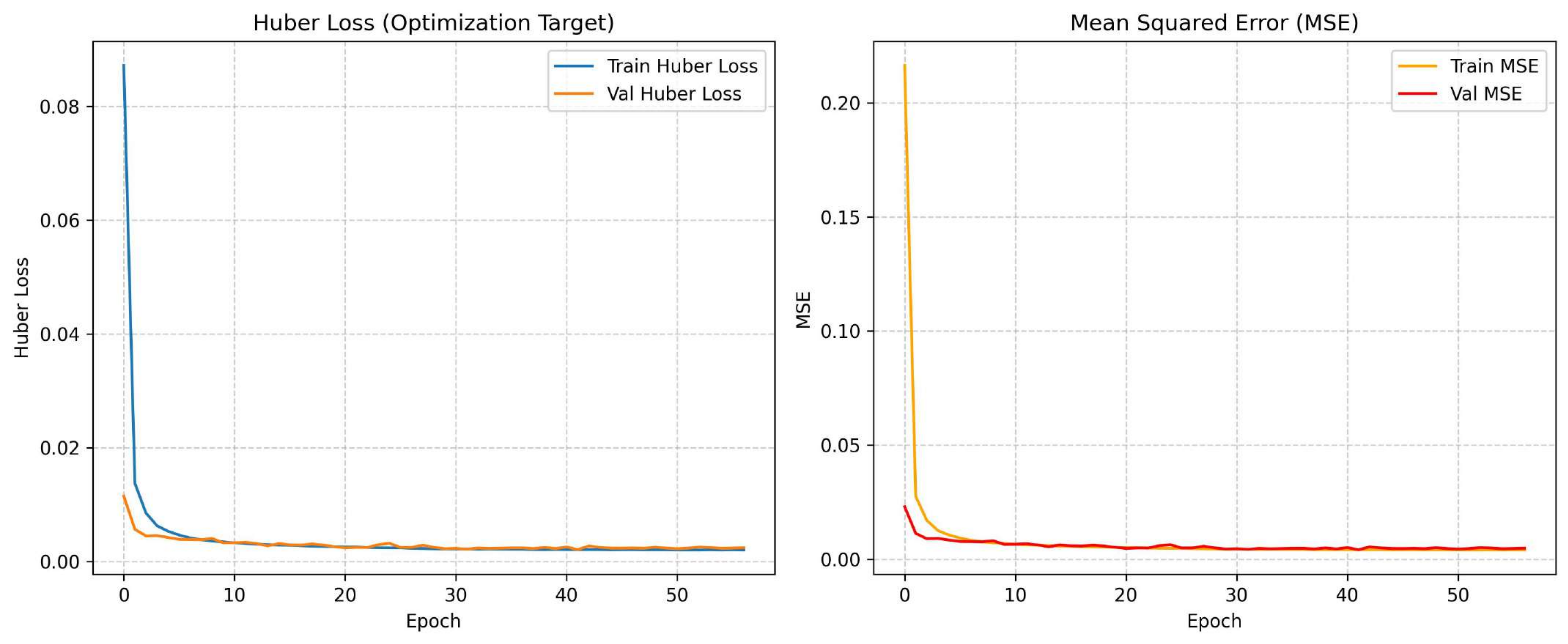
Metric	Training Value	Validation Value
Huber Loss	0.00205	0.00206
MSE	0.00409	0.00413
MAE	0.03598	0.03582

RESULTS (PRE and POST MODEL TUNING)

Previous ANN Model

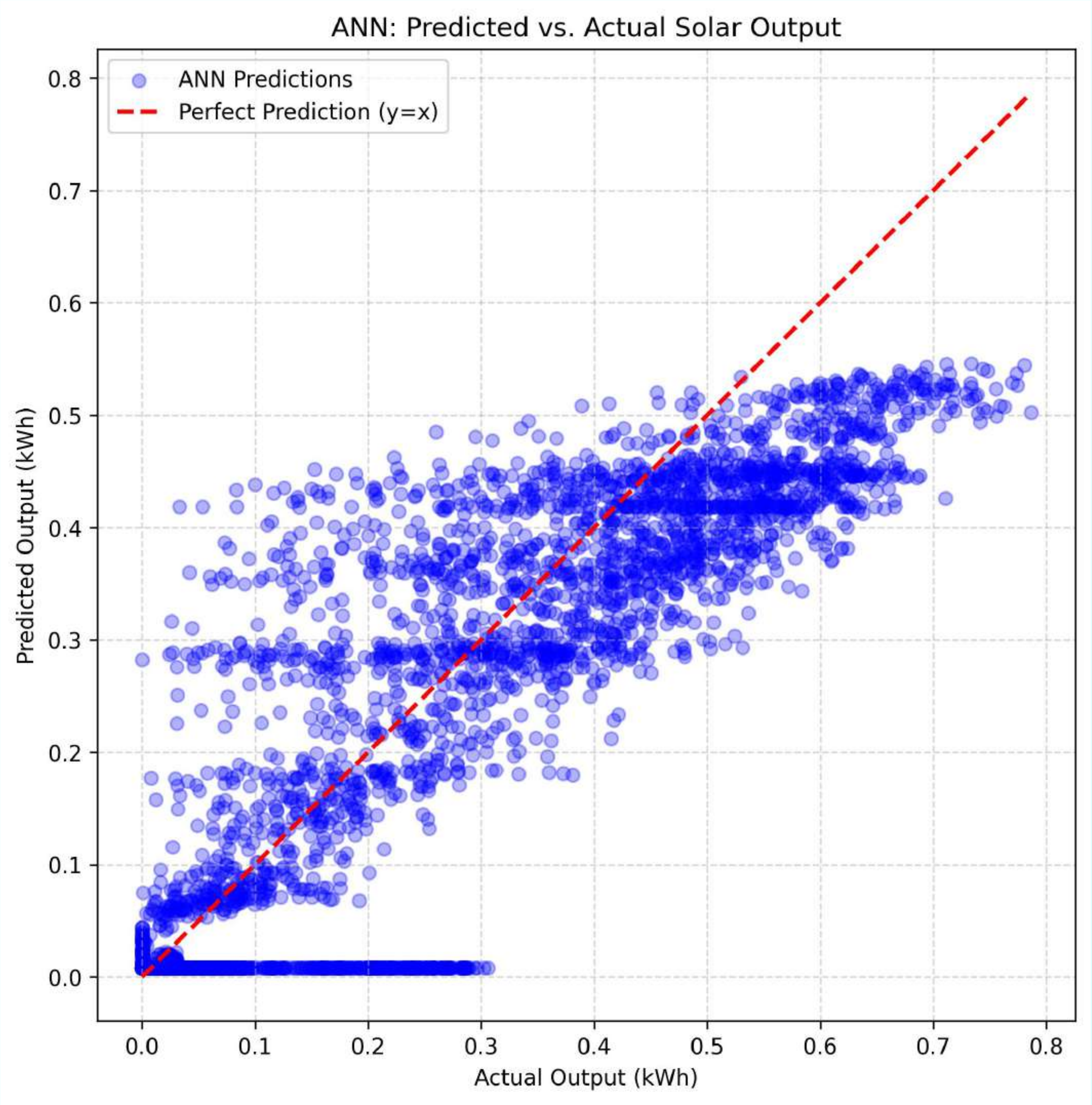


Current ANN Model

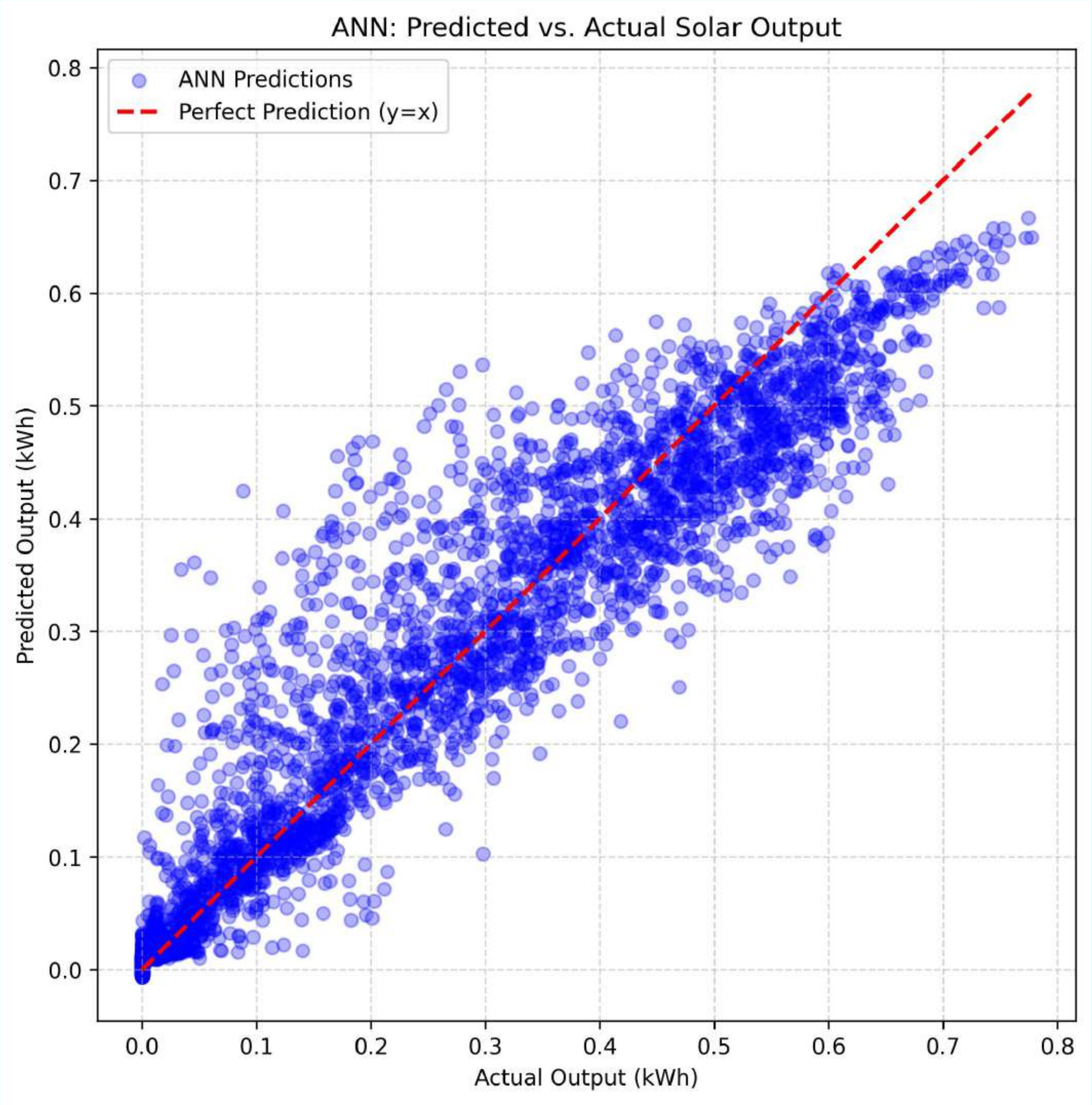


RESULTS (PREDICTED VS ACTUAL SCATTER PLOT)

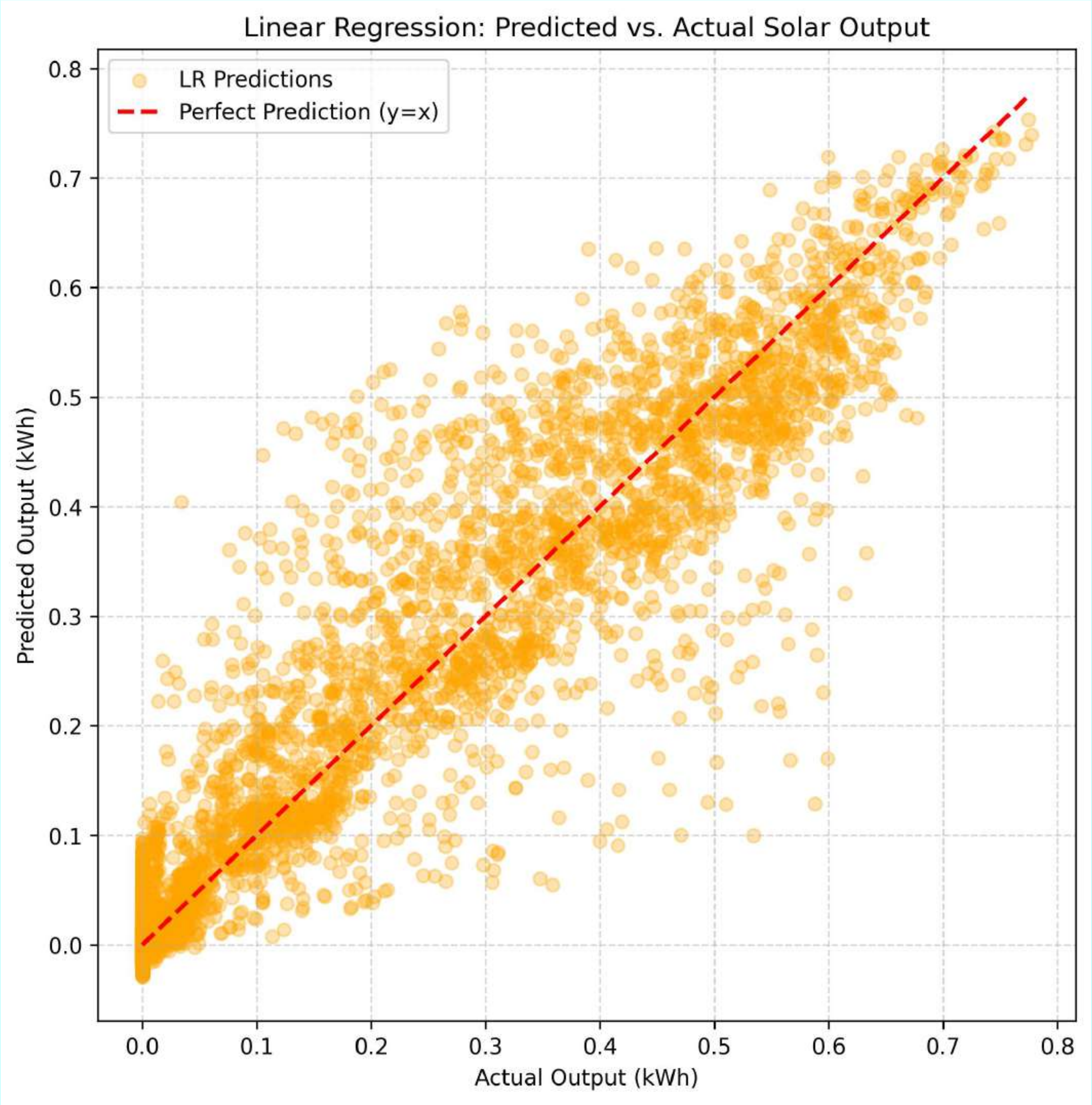
Previous ANN Model



Current ANN Model

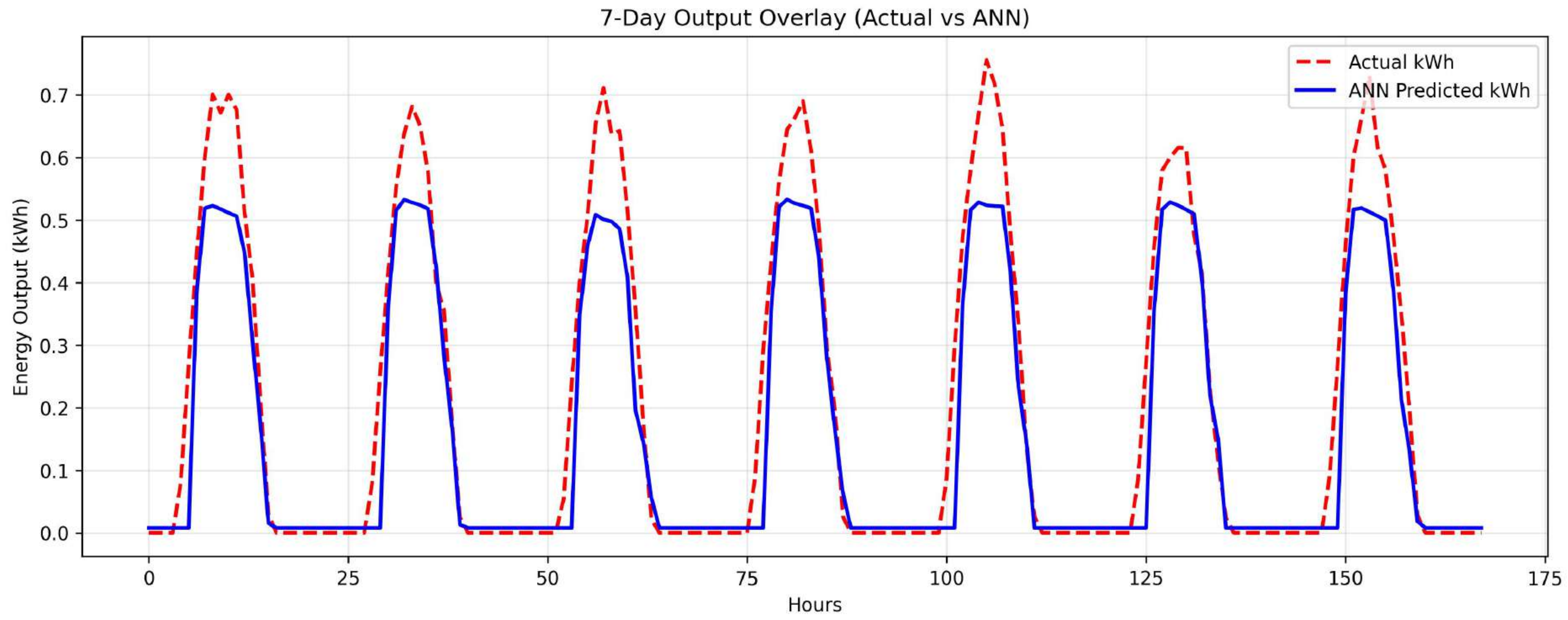


Linear Regression (Baseline)

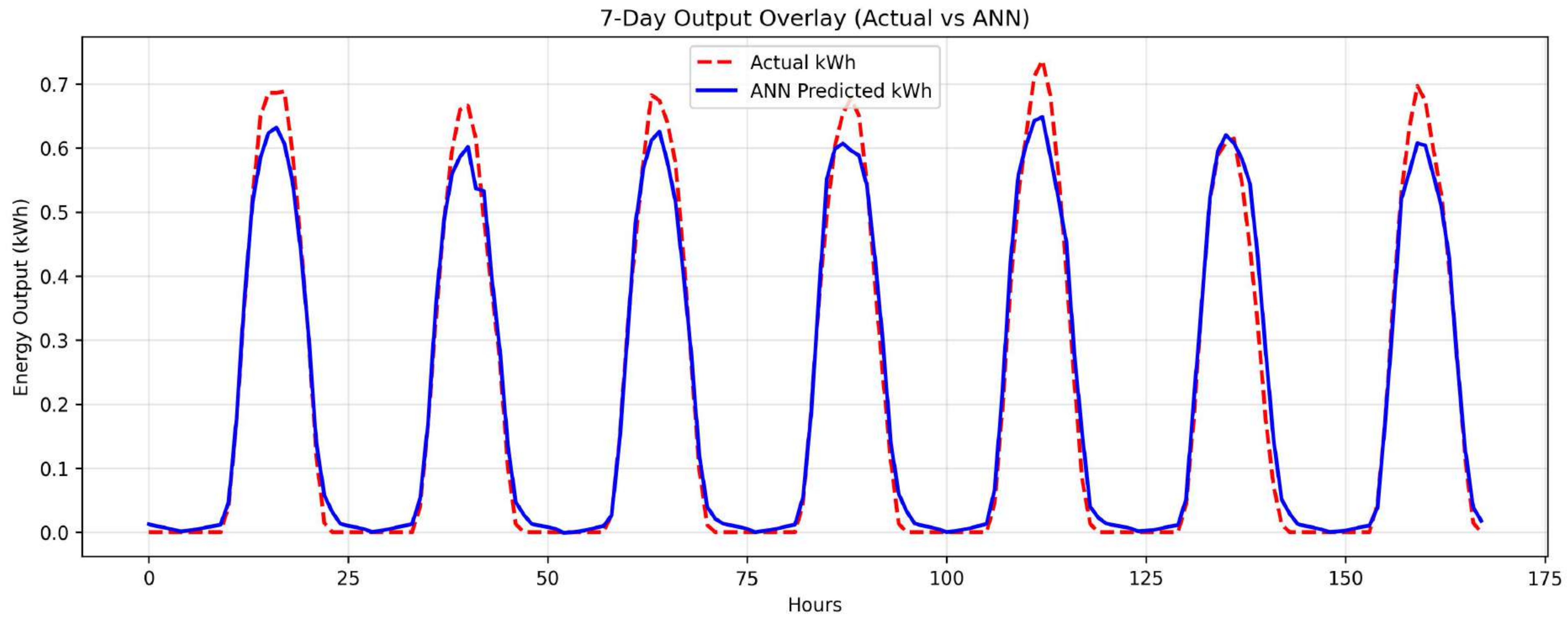


RESULTS (PREDICTED VS ACTUAL 7-DAY OUTPUT)

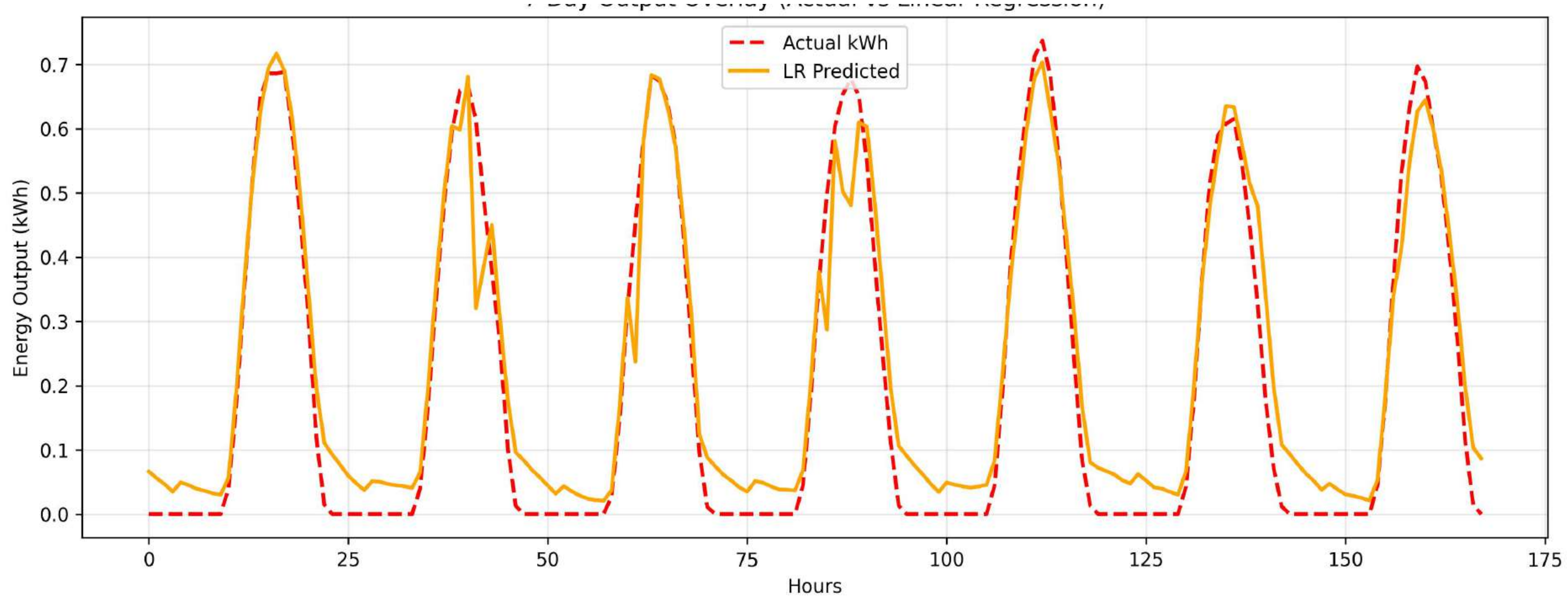
Previous ANN Model



Current ANN Model

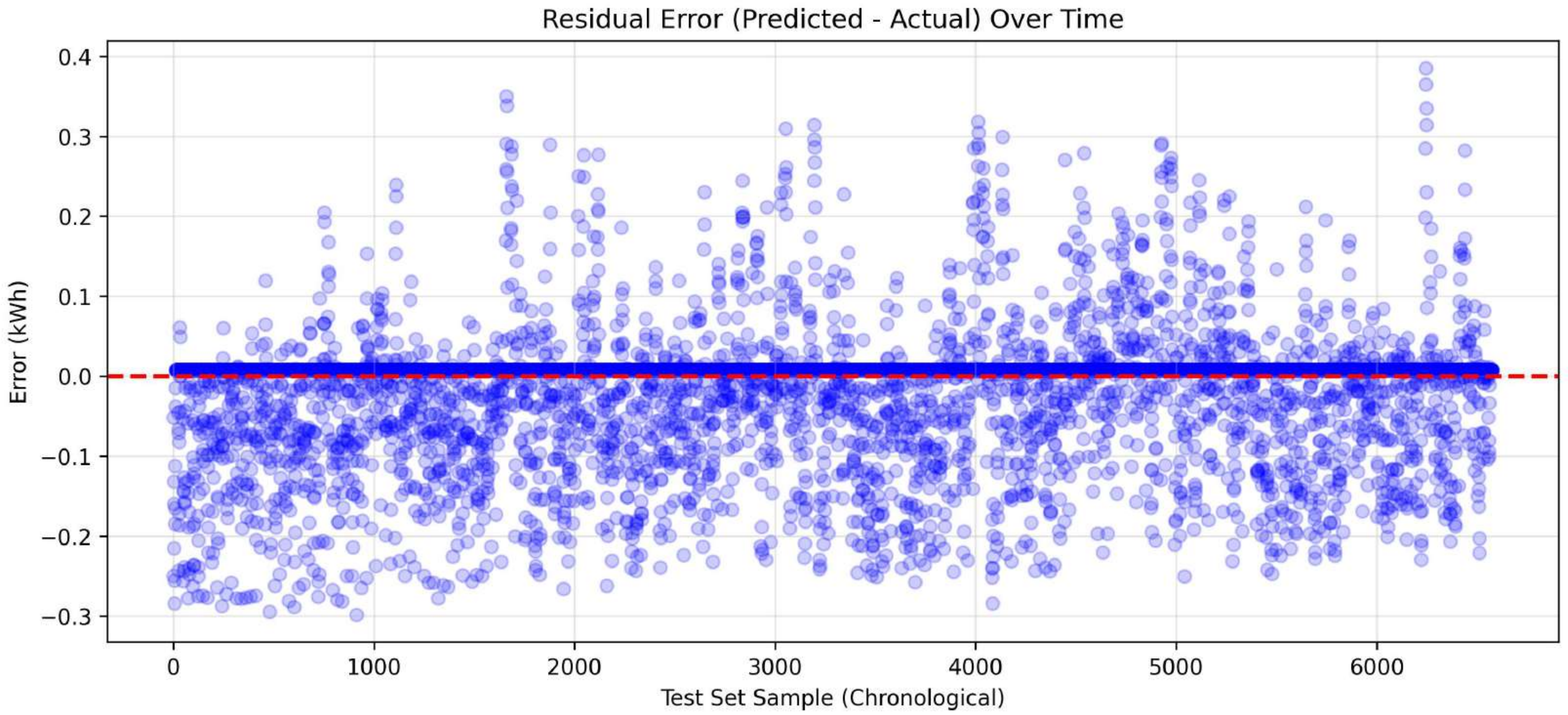


Linear Regression

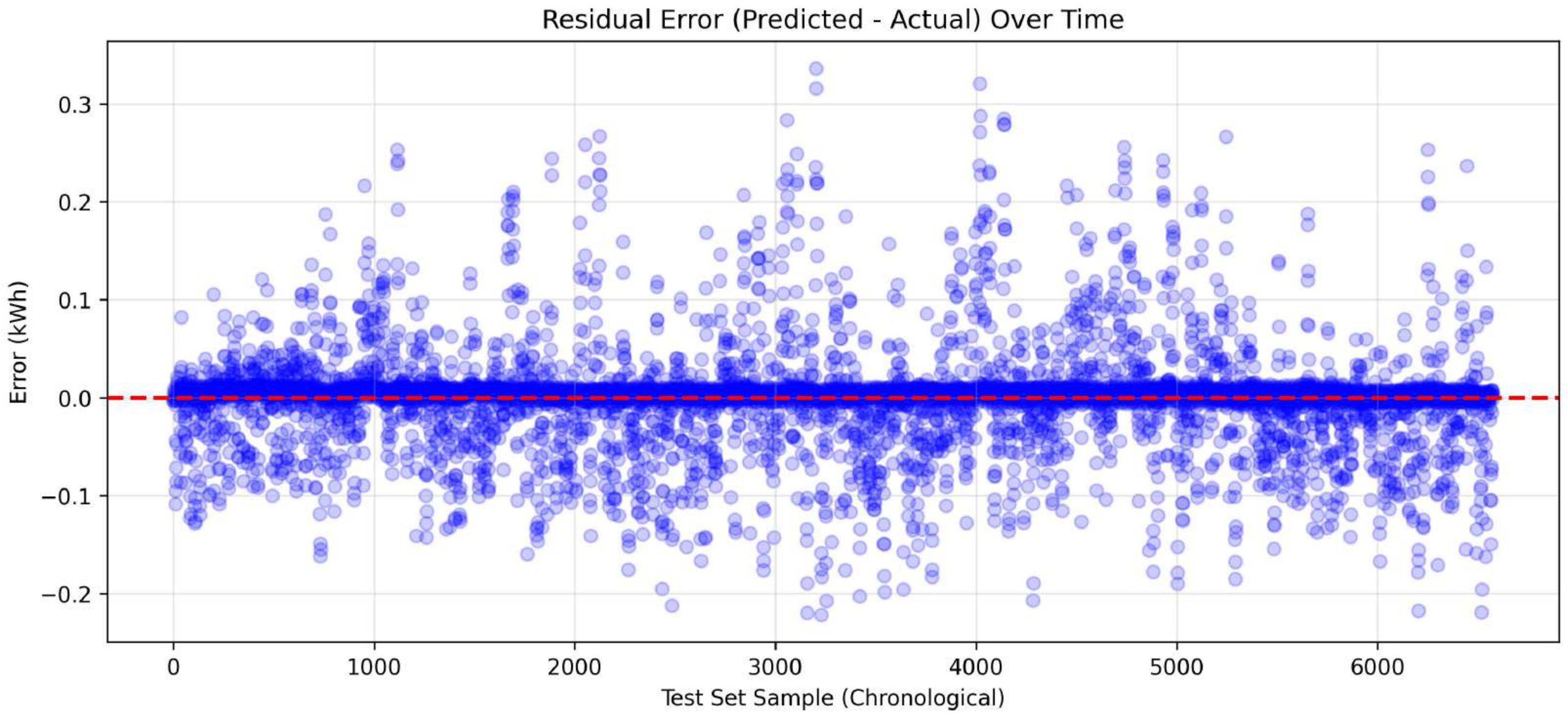


RESULTS (PREDICTED VS ACTUAL RESIDUAL ERROR)

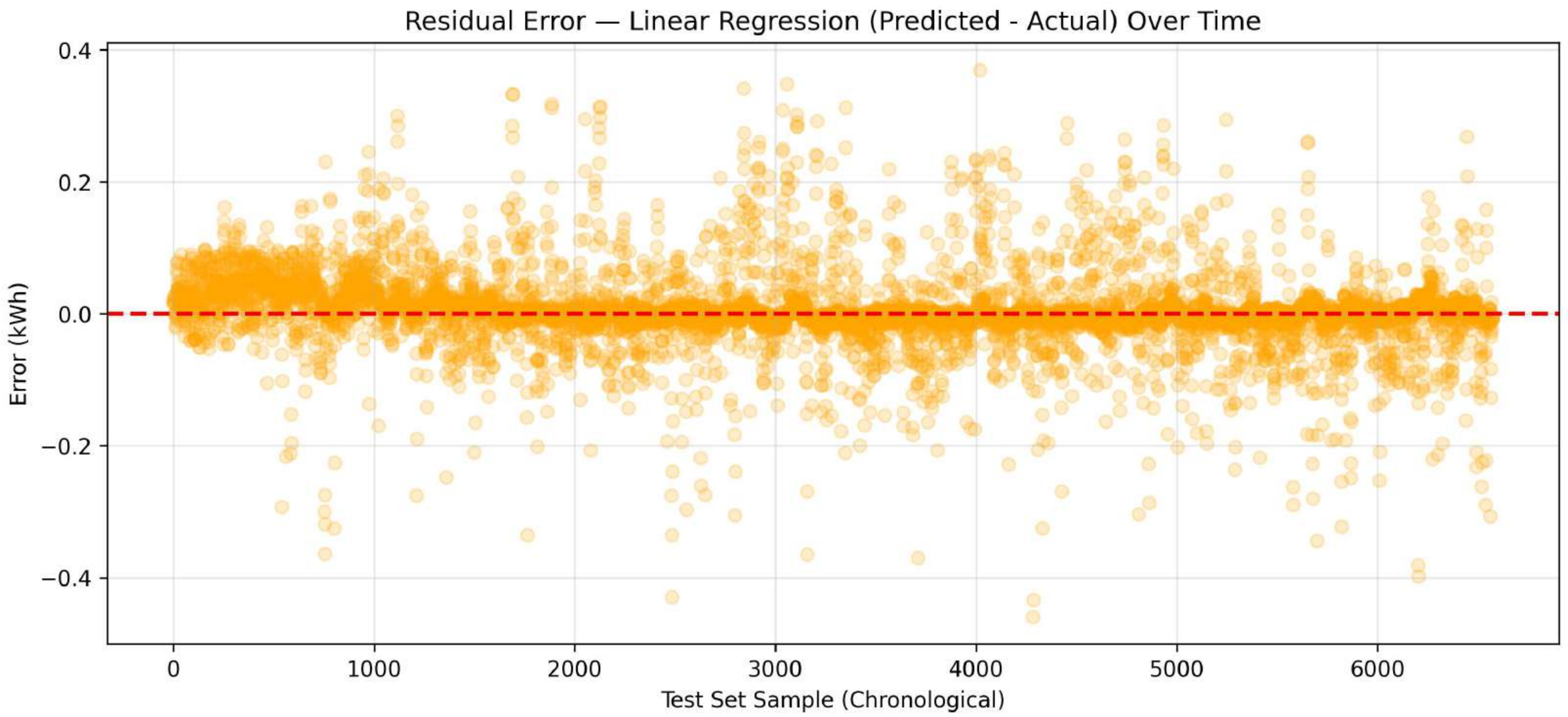
Previous ANN Model



Current ANN Model



Linear Regression



RESULTS (MODEL PERFORMANCE)

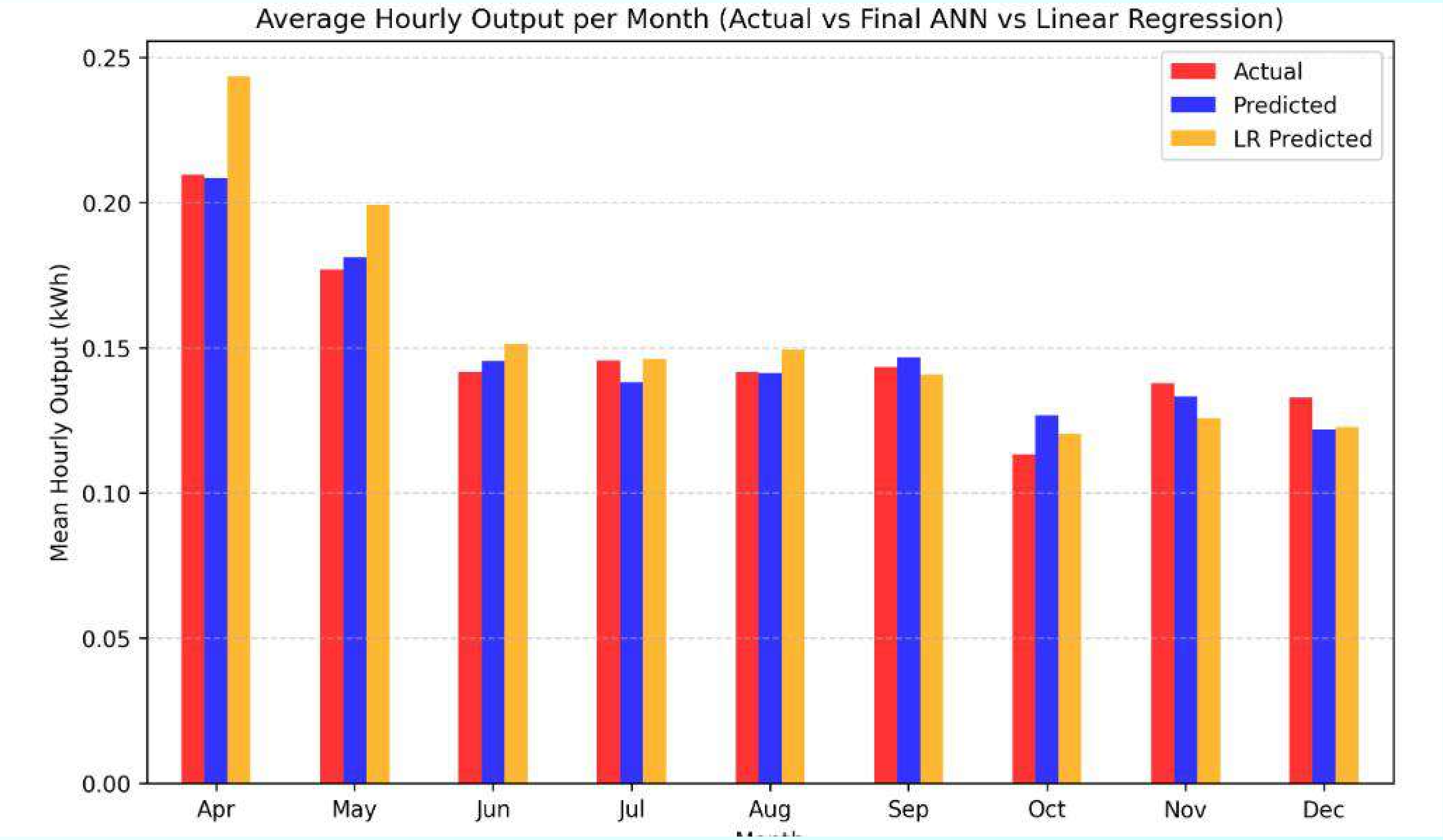
19.6%
RMSE Improvement
vs. Linear Regression

+0.0354
R^2 Boost
vs. Linear Regression

24.0%
MAE Improvement
vs. Linear Regression

Model	RMSE (kWh)	MAE (kWh)	R^2
Previous ANN Model	N/A	N/A	N/A
Current ANN Model	0.0519	0.0291	0.9356
Linear Regression Baseline	0.0646	0.0383	0.9002

FINDINGS & DISCUSSION



STRENGTHS

- $R^2 = 0.9356$ — explains 93.5% of PV output variance
- LeakyReLU resolved 'dying ReLU' during zero-output nighttime hours
- Huber loss reduced sensitivity to wet-season outliers
- Near-identical train/val metrics confirm no overfitting
- GHI ($r = 0.947$) dominates predictions, as physically expected

LIMITATIONS

- Accuracy dips during wet season (Jun–Nov typhoon cloud cover)
- MLP treats each hour independently — no temporal memory
- Optimized for Iloilo City only; not generalized across PH
- Cloud type features excluded despite availability in NSRDB